Quantum Interference in Quantum Computing: A Mathematical Introduction

Introduction

Quantum interference is one of the most fundamental phenomena that distinguishes quantum computation from classical computation. It refers to the way probability amplitudes—complex numbers associated with quantum states—combine according to the rules of wave mechanics: they can *constructively interfere* (reinforcing each other) or *destructively interfere* (canceling each other out).

In quantum algorithms, interference is carefully engineered through sequences of quantum gates to amplify the probability of correct answers while suppressing incorrect ones. This document presents a mathematical introduction to quantum interference using simple one- and two-qubit examples, and then connects these ideas to Grover's search algorithm.

Classical vs. Quantum Probability

In classical probability theory, probabilities are non-negative real numbers that add linearly. If an event can occur via two mutually exclusive paths with probabilities p_1 and p_2, the total probability is

$$P_{\text{total}} = p_1 + p_2. \tag{1}$$

In quantum mechanics, each computational path is associated with a *complex probability amplitude* $\alpha \in \mathbb{C}$. The probability of an outcome is given by the squared modulus of the *sum* of amplitudes:

$$P_{\text{quantum}} = |\alpha_1 + \alpha_2|^2. \tag{2}$$

Because amplitudes are complex, they have both magnitude and phase. When two amplitudes are added, their phases determine whether they reinforce (constructive interference) or cancel (destructive interference). This is fundamentally different from classical probability addition.

Mathematical Preliminaries

Qubit States and Superposition

A single qubit lives in a two-dimensional Hilbert space spanned by the computational basis states $|0\rangle$ and $|1\rangle$. A general qubit state is a *superposition*:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \tag{3}$$

where $\alpha, \beta \in \mathbb{C}$ satisfy the normalization condition

$$|\alpha|^2 + |\beta|^2 = 1. \tag{4}$$

Upon measurement in the computational basis, the qubit collapses to $|0\rangle$ with probability $|\alpha|^2$ and to $|1\rangle$ with probability $|\beta|^2$.

The Hadamard Gate

The Hadamard gate H is defined by

$$H\,|0\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big) \equiv |+\rangle\,, \qquad H\,|1\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle - |1\rangle\big) \equiv |-\rangle\,. \tag{5}$$

In matrix form,

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad H^2 = I. \tag{6}$$

Example 1: Interference in the H–H Circuit

Consider the circuit that applies two Hadamard gates in succession to a qubit initialized in $|0\rangle$.

Step 1: First Hadamard creates superposition

From $|0\rangle$, the first Hadamard gate produces

$$|\psi_1\rangle = H\,|0\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big) = |+\rangle\,. \tag{7}$$

At this stage, the qubit is in an equal superposition of both computational basis states. If we measured now, we would obtain $|0\rangle$ or $|1\rangle$ each with probability $\frac{1}{2}$.

Step 2: Second Hadamard causes interference

We now apply a second Hadamard:

$$\begin{aligned}
|\psi_2\rangle = H\,|+\rangle &= H\left(\frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)\right) && (8)\\
&= \frac{1}{\sqrt{2}}\big(H\,|0\rangle + H\,|1\rangle\big) && (9)\\
&= \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big) + \frac{1}{\sqrt{2}}\big(|0\rangle - |1\rangle\big)\right) && (10)\\
&= \frac{1}{2}\Big[\big(|0\rangle + |1\rangle\big) + \big(|0\rangle - |1\rangle\big)\Big] && (11)\\
&= \frac{1}{2}\big(2\,|0\rangle\big) = |0\rangle\,. && (12)
\end{aligned}$$

Thus the final state is $|0\rangle$ with certainty. The amplitudes for $|1\rangle$ have interfered destructively.

Path Interpretation

After the first Hadamard, the qubit is in

$$\frac{1}{\sqrt{2}}\,|0\rangle + \frac{1}{\sqrt{2}}\,|1\rangle\,.$$

When the second Hadamard is applied, the contribution to $|1\rangle$ from the "path via $|0\rangle$" is $+\frac{1}{2}$, while the contribution from the "path via $|1\rangle$" is $-\frac{1}{2}$. These cancel, giving zero total amplitude for $|1\rangle$. This is destructive interference.

Example 2: Phase Flip and Interference Pattern

Consider the circuit $|0\rangle \xrightarrow{H}\xrightarrow{Z}\xrightarrow{H}$, where the Pauli-$Z$ gate is

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Z|0\rangle = |0\rangle\,, \quad Z|1\rangle = -|1\rangle\,. \tag{13}$$

Step-by-step evolution

First Hadamard:

$$|\psi_1\rangle = H|0\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big). \tag{14}$$

Phase flip via Z:

$$|\psi_2\rangle = Z|\psi_1\rangle = \frac{1}{\sqrt{2}}\big(Z|0\rangle + Z|1\rangle\big) \tag{15}$$

$$= \frac{1}{\sqrt{2}}\big(|0\rangle - |1\rangle\big) = |-\rangle\,. \tag{16}$$

Second Hadamard:

$$|\psi_3\rangle = H|-\rangle = H\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \tag{17}$$

$$= \frac{1}{\sqrt{2}}\big(H|0\rangle - H|1\rangle\big) \tag{18}$$

$$= \frac{1}{\sqrt{2}}\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} - \frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \tag{19}$$

$$= \frac{1}{2}\Big[(|0\rangle + |1\rangle) - (|0\rangle - |1\rangle)\Big] \tag{20}$$

$$= \frac{1}{2}(2|1\rangle) = |1\rangle\,. \tag{21}$$

The Z gate changed the relative phase, thus flipping the interference pattern: now $|1\rangle$ experiences constructive interference and $|0\rangle$ destructive interference.

Example 3: Two-Qubit Interference (Deutsch–Jozsa with $n = 2$)

We now consider a simple two-qubit example that generalises the one-qubit interference picture and is closely related to the Deutsch–Jozsa algorithm.

Setup

Consider two input qubits and one ancilla qubit, all initialised as

$$|\psi_0\rangle = |00\rangle\,|1\rangle\,. \tag{22}$$

We will:

1. Apply Hadamards to all three qubits.
2. Apply an oracle U_f that encodes a Boolean function $f : \{0,1\}^2 \to \{0,1\}$.
3. Apply Hadamards again to the two input qubits.
4. Measure the input register.

The oracle is defined by

$$U_f\,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus f(x)\rangle\,, \tag{23}$$

where $x \in \{0,1\}^2$ and $\oplus$ is XOR.

Step 1: Create uniform superposition

Applying $H^{\otimes 3}$ to $|00\rangle\,|1\rangle$ yields

$$|\psi_1\rangle = (H^{\otimes 2}\,|00\rangle) \otimes (H\,|1\rangle) \tag{24}$$

$$= \left(\frac{1}{2}\sum_{x\in\{0,1\}^2} |x\rangle\right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \tag{25}$$

$$= \frac{1}{2\sqrt{2}}\sum_{x\in\{0,1\}^2} |x\rangle\,\big(|0\rangle - |1\rangle\big). \tag{26}$$

Thus each input basis state $|x\rangle$ appears with equal amplitude, and the ancilla is in the $|-\rangle$ state.

Step 2: Apply the oracle U_f

We act with U_f:

$$|\psi_2\rangle = \frac{1}{2\sqrt{2}}\sum_{x\in\{0,1\}^2} U_f\Big(|x\rangle\,\big(|0\rangle - |1\rangle\big)\Big) \tag{27}$$

$$= \frac{1}{2\sqrt{2}}\sum_{x} \left(|x\rangle\,|0\oplus f(x)\rangle - |x\rangle\,|1\oplus f(x)\rangle\right). \tag{28}$$

We use the fact that

$$|0\oplus f(x)\rangle - |1\oplus f(x)\rangle = (-1)^{f(x)}\big(|0\rangle - |1\rangle\big),$$

so

$$|\psi_2\rangle = \frac{1}{2\sqrt{2}}\sum_{x}(-1)^{f(x)}\,|x\rangle\,\big(|0\rangle - |1\rangle\big) \tag{29}$$

$$= \left(\frac{1}{2}\sum_{x}(-1)^{f(x)}\,|x\rangle\right) \otimes |-\rangle\,. \tag{30}$$

The ancilla returns to $|-\rangle$ and all information about f is now encoded in the phases $(-1)^{f(x)}$ of the input superposition. This is a phase kickback effect.

Step 3: Final Hadamards and interference

We now apply $H^{\otimes 2}$ to the input qubits:

$$|\psi_3\rangle = \left(H^{\otimes 2}\frac{1}{2}\sum_{x}(-1)^{f(x)}\,|x\rangle\right) \otimes |-\rangle\,. \tag{31}$$

Using

$$H^{\otimes 2}\,|x\rangle = \frac{1}{2}\sum_{z\in\{0,1\}^2}(-1)^{x\cdot z}\,|z\rangle\,,$$

where $x\cdot z$ is the bitwise inner product modulo 2, we obtain

$$|\psi_3\rangle = \frac{1}{4}\sum_{x,z}(-1)^{f(x)}(-1)^{x\cdot z}\,|z\rangle \otimes |-\rangle\,. \tag{32}$$

The amplitude of each output basis state $|z\rangle$ is

$$\alpha_z = \frac{1}{4}\sum_{x}(-1)^{f(x)+x\cdot z}.$$

Constant f. If f is constant (always 0 or always 1), the phase $(-1)^{f(x)}$ is the same for all x. In that case, for $z = 00$ every term in the sum has the same sign and adds constructively, while for $z \neq 00$ the terms cancel pairwise. Thus the input register ends up in $|00\rangle$ with certainty.

Balanced f. If f is balanced (half of the inputs map to 0 and half to 1), the sum for $z = 00$ splits into two equal halves with opposite signs and cancels, so the amplitude of $|00\rangle$ is zero. The final state has no support on $|00\rangle$; interference has removed it completely.

This two-qubit example shows how phase information introduced by the oracle is converted, via interference, into observable differences in measurement outcomes.

Interference in Grover's Algorithm

We now briefly describe how interference appears in Grover's search algorithm through the *diffusion operator*.

Uniform superposition and marking

Consider an n-qubit register with computational basis $\{|x\rangle\}_{x\in\{0,1\}^n}$. Start from

$$|\psi_0\rangle = |0\rangle^{\otimes n}, \tag{33}$$

then apply $H^{\otimes n}$ to obtain the uniform superposition

$$|s\rangle = H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{N}}\sum_x |x\rangle, \qquad N = 2^n. \tag{34}$$

Suppose one basis state $|w\rangle$ is "marked". The oracle O acts as

$$O|x\rangle = \begin{cases} -|x\rangle & \text{if } x = w, \\ |x\rangle & \text{otherwise.} \end{cases} \tag{35}$$

That is, the oracle flips the phase of the marked state but leaves all other states unchanged.

Diffusion operator as inversion about the average

Grover's diffusion operator D is defined as

$$D = 2|s\rangle\langle s| - I. \tag{36}$$

To see its action on a general state

$$|\psi\rangle = \sum_x a_x |x\rangle,$$

observe that

$$\langle s|\psi\rangle = \frac{1}{\sqrt{N}}\sum_x a_x = \sqrt{N}\,\bar{a},$$

where $\bar{a}$ is the average amplitude. Then

$$\begin{aligned} D|\psi\rangle &= \big(2|s\rangle\langle s| - I\big)|\psi\rangle && (37)\\ &= 2|s\rangle\langle s|\psi\rangle - |\psi\rangle && (38)\\ &= 2|s\rangle\big(\sqrt{N}\,\bar{a}\big) - \sum_x a_x|x\rangle && (39)\\ &= 2\bar{a}\sum_x |x\rangle - \sum_x a_x|x\rangle && (40)\\ &= \sum_x (2\bar{a} - a_x)|x\rangle. && (41) \end{aligned}$$

Thus each amplitude a_x is mapped to $2\bar{a} - a_x$: reflection about the mean.

One Grover iteration as interference

A single Grover iteration applies $G = DO$ to the current state.

- The oracle O flips the sign of the marked amplitude a_w, leaving all others unchanged. This is a phase flip, analogous to the Z gate in the one-qubit example.
- The diffusion operator D reflects all amplitudes about their average, which now differs from before because of the sign flip.

The combined effect is:

- The marked state's amplitude is *increased* in magnitude (constructive interference).
- The unmarked states' amplitudes are slightly *decreased* (destructive interference relative to the mean).

Repeating G approximately $\frac{\pi}{4}\sqrt{N}$ times rotates the state vector in the two-dimensional subspace spanned by $|w\rangle$ and the uniform superposition of unmarked states, bringing it close to $|w\rangle$. Measurement then yields the marked element with high probability. The speedup arises entirely from carefully controlled interference between the amplitudes.

Why Interference Matters

Quantum interference is what enables quantum computers to perform certain computations exponentially or quadratically faster than classical computers. Classical machines must evaluate each possibility separately and combine probabilities. Quantum computers evaluate many possibilities in superposition, then use interference to sculpt the probability landscape so that correct answers are amplified and incorrect answers are suppressed.

Interference requires the quantum system to maintain coherence. External noise and uncontrolled interactions cause decoherence, which destroys the delicate interference patterns. Building scalable, fault-tolerant quantum computers therefore requires keeping coherence long enough and using error correction to protect the interference structure that algorithms rely on.